

1 DATA

WildChat-1M corpus

- 509,861 conversations
- Apr 2023 – Apr 2024 (386 days)
- Role + content per turn

Cohort filter

- ≥ 10 sessions per user retained
- Redacted conversations excluded (0.9%)
- 505,397 sessions in analysis

12,577 qualifying users

2 FEATURES

Lexical

- FPP rate | Hedging | Self-disclosure
- Filler / phatic expression rate

Pragmatic

- Refusal resistance rate
- Gratitude rate | Q–S ratio

Session-level

- Type–token ratio | Message length
- Return interval | Sentiment slope

12 features · single-pass token scan

3 WINDOWS & LABEL

Sessions 1–3

Sessions 10+



Dependency label (≥ 2 of 4)

- Return interval $<$ cohort median
- Session length $>$ 75th percentile
- Refusal resistance rate > 0
- FPP rate $>$ cohort mean (late sessions)

High-dep: 24.1% (n=3,014 / 12,497)

4 MODELLING

Univariate baselines

- Logistic regression per feature
- AUC-ROC + Cohen's d
- BH FDR correction (12 tests)

Main model

- L1-regularised logistic regression
- User-level stratified 10-fold CV
- class_weight=balanced

AUC=0.712 $\pm$ 0.014

5 VALIDATION

Permutation test

- 1,000 label shuffles
- $p < 0.001$ (0 / 1,000 exceed observed)

Negative control

- Punctuation rate (theoretically null)
- Rank: 14 / 15 · coef ≈ 0 [pass]

Sensitivity analysis

- Threshold ≥ 1 – ≥ 3 | AUC: 0.69–0.75

Non-spurious · Neg. control: PASSED

AUC=0.712 $\pm$ 0.014

Permutation $p < 0.001$

9/12 features BH-significant

Negative control rank 14/15

Feature ranking stable across thresholds